

Algebra Factoring using a Single Parenthesis



Factoring into single parentheses – Part 1

1) Each of the following has 2 as a common factor.

a) $2x + 4 = 2(\quad + \quad)$

c) $2x + 6 = 2(\quad + \quad)$

b) $2x - 12 = 2(\quad - \quad)$

d) $4x - 14 = 2(\quad - \quad)$

2) Each of the following has 5 as a common factor.

a) $5x + 20 = 5($

c) $5x + 15 = 5($

b) $10x - 15 = 5($

d) $40x - 15 = 5($

3) Each of the following has 4 as a common factor.

a) $4x + 20 = 4($

c) $8x + 16 = 4($

b) $16x - 20 = 4($

d) $40x - 12 = 4($

4) Each of the following has 7 as a common factor.

a) $7x + 21 = 7($

c) $7x + 14 = 7($

b) $14x - 21 = 7($

d) $42x - 21 = 7($

5) Each of the following has 10 as a common factor.

a) $10x + 20 = 10($

c) $70x + 10 = 10($

b) $40x - 50 = 10($

d) $30x - 20 = 10($



Factoring into single parentheses – Part 2

Draw lines to match each algebra expression with its factorized form.

$4x + 8$

$7(x - 2)$

$3x + 6$

$10(x + 3)$

$5x + 15$

$8(x - 2)$

$7x - 14$

$4(x + 2)$

$6x - 30$

$9(x - 3)$

$10x + 30$

$6(x - 5)$

$2x + 8$

$3(x - 4)$

$8x - 16$

$3(x + 2)$

$3x - 12$

$2(x + 4)$

$9x - 27$

$5(x + 3)$



Factoring into single parentheses – Part 3

Factor the following expressions.

1) $2x + 8 = 2($

17) $4x - 12$

2) $3x + 9 = 3($

18) $21 + 3x =$

3) $2x + 6 = 2($

19) $40 - 4x =$

4) $5x + 20 = 5($

20) $63 + 7x =$

5) $2x + 10 = 2($

21) $12 - 8y =$

6) $6x + 3 = 3($

22) $14 + 2b =$

7) $7x + 7 = 7($

23) $20 + 15d =$

8) $11x + 22 = 11($

24) $56 - 12a =$

9) $9x + 18 = 9($

25) $18 + 12c =$

10) $4x + 36 = 4($

26) $22 - 33f =$

11) $3x + 21 =$

27) $6 + 15x =$

12) $4a + 16 =$

28) $3x - 12 =$

13) $5y + 10 =$

29) $5x - 25 =$

14) $7y + 35 =$

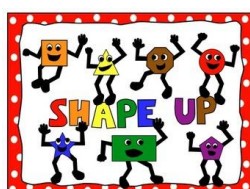
30) $8x - 10 =$

15) $12b + 6 =$

31) $70x - 14 =$

16) $2g + 4 =$

32) $14x - 21 =$



Factoring into single parentheses – Part 4

Factor the following expressions.

1) $6y - 12 =$

2) $56 - 42x =$

3) $21 - 28x =$

4) $50r - 10 =$

5) $30t - 40 =$

6) $10p - 25 =$

7) $2a - 34 =$

8) $36 - 12t =$

9) $72 - 18y =$

10) $100 - 10y =$

11) $24x + 32 =$

12) $15x + 18 =$

13) $55x + 5 =$

14) $45y + 27 =$

15) $81x - 90 =$

16) $99x + 77 =$

17) $x^2 + 3x = x($

18) $x^2 + 7x = x($

19) $a^2 - 8a =$

20) $x^2 - 4x =$

21) $2x^2 + 4x = 2x($

22) $3y^2 + 3y = 3y($

23) $2x^2 + 6x =$

24) $2x^2 - 8x =$

25) $5x^2 + 30x =$

26) $10p^2 - 15p =$

27) $36x^2 + 42x =$

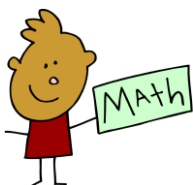
28) $10x^2 + 30x =$

29) $25t^2 + 50t =$

30) $8x^2 - 12x =$

31) $7x^2 + 35x =$

32) $18n^2 - 27n =$





Factoring Test

Factor the following expressions.

1) $4x + 10$

2) $6x - 8$

3) $5x + 10$

4) $3x + 12$

5) $14x + 21$

6) $x^2 + 4x$

7) $x^2 + 10x$

8) $x^2 - 6x$

9) $2x^2 + 8x$

10) $10x^2 + 25x$

ANSWERS - Factoring into single parentheses – Part 1

1) Each of the following has 2 as a common factor.

a) $2x + 4 = 2(x + 2)$

c) $2x + 6 = 2(x + 3)$

b) $2x - 12 = 2(x - 6)$

d) $4x - 14 = 2(2x - 7)$

2) Each of the following has 5 as a common factor.

a) $5x + 20 = 5(x+4)$

c) $5x + 15 = 5(x+3)$

b) $10x - 15 = 5(2x-3)$

d) $40x - 15 = 5(8x-3)$

3) Each of the following has 4 as a common factor.

a) $4x + 20 = 4(x+5)$

c) $8x + 16 = 4(2x+4)$

b) $16x - 20 = 4(4x-5)$

d) $40x - 12 = 4(10x-3)$

4) Each of the following has 7 as a common factor.

a) $7x + 21 = 7(x+3)$

c) $7x + 14 = 7(x+2)$

b) $14x - 21 = 7(2x-3)$

d) $42x - 21 = 7(6x - 3)$

5) Each of the following has 10 as a common factor.

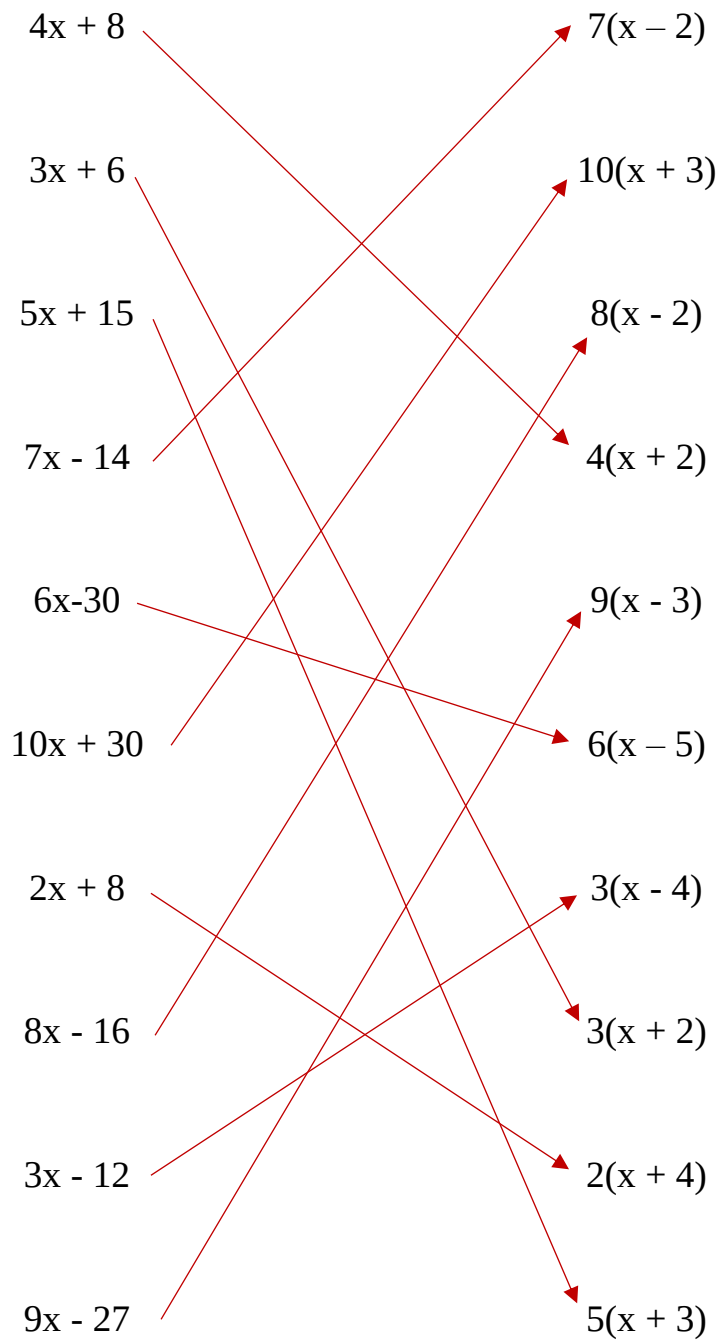
a) $10x + 20 = 10(x + 2)$

c) $70x + 10 = 10(7x + 1)$

b) $40x - 50 = 10(4x - 5)$

d) $30x - 20 = 10(3x - 2)$

ANSWERS - Factoring into single parentheses – Part 2



ANSWERS - Factoring into single parentheses – Part 3

1) $2x + 8 = 2(x+4)$

2) $3x + 9 = 3(x+3)$

3) $2x + 6 = 2(x+3)$

4) $5x + 20 = 5(x+4)$

5) $2x + 10 = 2(x+5)$

6) $6x + 3 = 3(2x+1)$

7) $7x + 7 = 7(x+1)$

8) $11x + 22 = 11(x+2)$

9) $9x + 18 = 9(x+2)$

10) $4x + 36 = 4(x+9)$

11) $3x + 21 = 3(x+7)$

12) $4a + 16 = 4(a+4)$

13) $5y + 10 = 5(y+2)$

14) $7y + 35 = 7(y+5)$

15) $12b + 6 = 6(2b + 1)$

16) $2g + 4 = 2(g+2)$

17) $4x - 12 = 4(x-3)$

18) $21 + 3x = 3(7+x)$

19) $40 - 4x = 4(10-x)$

20) $63 + 7x = 7(9+x)$

21) $12 - 8y = 4(3-2y)$

22) $14 + 2b = 2(7+b)$

23) $20 + 15d = 5(4+3d)$

24) $56 - 12a = 4(14-3a)$

25) $18 + 12c = 6(3+2c)$

26) $22 - 33f = 11(2-3f)$

27) $6 + 15x = 3(2+5x)$

28) $3x - 12 = 3(x-4)$

29) $5x - 25 = 5(x-5)$

30) $8x - 10 = 2(4x-5)$

31) $70x - 14 = 7(10x-2)$

32) $14x - 21 = 7(2x-3)$

ANSWERS - Factoring into single parentheses – Part 4

1) $6y - 12 = 6(y-2)$

2) $56 - 42x = 7(8-6x)$

3) $21 - 28x = 7(3-4x)$

4) $50r - 10 = 10(5r-1)$

5) $30t - 40 = 10(3t-4)$

6) $10p - 25 = 5(2p-5)$

7) $2a - 34 = 2(a-17)$

8) $36 - 12t = 12(3-t)$

9) $72 - 18y = 18(4-y)$

10) $100 - 10y = 10(10-y)$

11) $24x + 32 = 8(3x+4)$

12) $15x + 18 = 3(5x+6)$

13) $55x + 5 = 5(11x+1)$

14) $45y + 27 = 9(5y+3)$

15) $81x - 90 = 9(9x-10)$

16) $99x + 77 = 11(9x + 7)$

17) $x^2 + 3x = x(x+3)$

18) $x^2 + 7x = x(x+7)$

19) $a^2 - 8a = a(a-8)$

20) $x^2 - 4x = x(x-4)$

21) $2x^2 + 4x = 2x(x+2)$

22) $3y^2 + 3y = 3y(y+1)$

23) $2x^2 + 6x = 2x(x+3)$

24) $2x^2 - 8x = 2x(x-4)$

25) $5x^2 + 30x = 5x(x+6)$

26) $10p^2 - 15p = 5p(2p-3)$

27) $36x^2 + 42x = 6x(6x+7)$

28) $10x^2 + 30x = 10x(x+3)$

29) $25t^2 + 50t = 25t(t+2)$

30) $8x^2 - 12x = 4x(2x-3)$

31) $7x^2 + 35x = 7x(x+5)$

32) $18n^2 - 27n = 9n(2n-3)$

ANSWERS – Factoring Test

1) $4x + 10 = 2(2x + 5)$

2) $6x - 8 = 2(3x - 4)$

3) $5x + 10 = 5(x + 2)$

4) $3x + 12 = 3(x + 4)$

5) $14x + 21 = 7(2x + 3)$

6) $x^2 + 4x = x(x + 4)$

7) $x^2 + 10x = x(x + 10)$

8) $x^2 - 6x = x(x - 6)$

9) $2x^2 + 8x = 2x(x + 4)$

10) $10x^2 + 25x = 5x(2x + 5)$

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